NAME-ARBON ROLLNO.231030272 BATCH-A12 PROJECT NAME:RAILWAY NETWORKING SIMUATION

```
!pip install simpy
```

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Requirement already satisfied: simpy in /usr/local/lib/python3.11/dist-packages (4.1.1)
```

```
import simpy
import networkx as nx
import matplotlib.pyplot as plt
import random

# Create railway network using NetworkX
G = nx.Graph()
stations = ['A', 'B', 'C', 'D']
edges = [('A', 'B'), ('B', 'C'), ('C', 'D'), ('A', 'D')]
G.add_edges_from(edges)

# Display the network
nx.draw(G, with_labels=True, node_color='lightblue', node_size=1000, font_size=14)
plt.title("Railway Network")
plt.show()

# Train class
class Train:
    def __init__(self, env, name, route, travel_time):
        self.env = env
        self.name = name
        self.route = route
        self.travel_time = travel_time
        self.action = env.process(self.run())

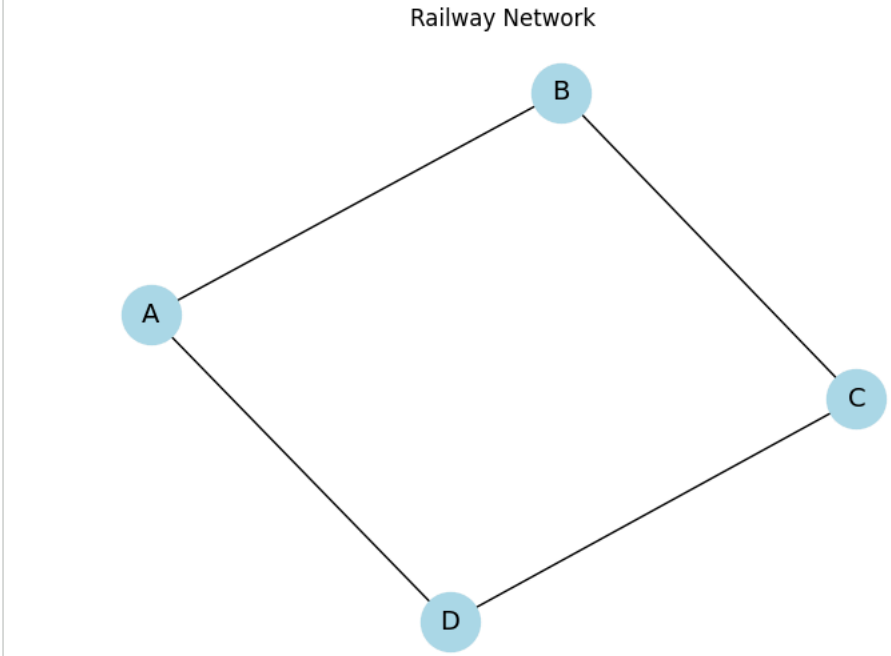
    def run(self):
        for i in range(len(self.route)-1):
            from_station = self.route[i]
            to_station = self.route[i+1]
            print(f"{self.name} departing from {from_station} to {to_station} at time {self.env.now}")
            yield self.env.timeout(self.travel_time)
            print(f"{self.name} arrived at {to_station} at time {self.env.now}")

# Simulation environment
env = simpy.Environment()

# Define train schedules
routes = [['A', 'B', 'C'], ['D', 'C', 'B'], ['A', 'D']]
travel_time = 3  # fixed travel time between stations

# Create trains
for i, route in enumerate(routes):
    Train(env, f"Train {i+1}", route, travel_time)

# Run the simulation
env.run()
```




```
Train 1 departing from A to B at time 0
Train 2 departing from D to C at time 0
Train 3 departing from A to D at time 0
Train 1 arrived at B at time 3
Train 1 departing from B to C at time 3
Train 2 arrived at C at time 3
Train 2 departing from C to B at time 3
```